

Paper B-12: Engineering Synthesis: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper closes the Part B engineered-systems sequence. It treats infrastructure, electrical grids, routing networks, distributed computation, cloud systems, knowledge graphs, autonomous systems, and computational architecture as engineering examples of the same CDFD/AFL modelling form. The claim is not that these systems share one material mechanism. The narrower claim is that throughput Φ , constraint C , responsiveness S , and structural memory M_s provide a common audit language for overload, backlog, fragility, and recovery. Release-local runtime outputs are model diagnostics and tests that could break the mapping, not engineering certification.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow-constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine [5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, is the

release archive that tests the notation across domains. Part B applies the shared notation to engineered systems, where every claim still needs ordinary engineering measurement.

2 Purpose

Part B translates Adaptive Flux Limitation (AFL) into engineered systems. The sequence begins with physical infrastructure and electrical grids, moves through routing, distributed computation, cloud systems, knowledge graphs, and autonomous systems, then ends with hardware architecture. This synthesis keeps the public release boundary clear: CDFD/AFL supplies a modelling vocabulary, not operational design authority.

3 Common Engineering Map

CDFD term	Engineering reading	Example observable
Φ	throughput or demand drive	current, packets, jobs, vehicles, queries
C	active constraint	line capacity, latency, lock contention, thermal budget
S	responsiveness	rerouting, autoscaling, buffering, adaptive control
M_s	structural memory	legacy topology, statefulness, cached assumptions
Ψ_s	operating ratio	overload, backlog, underuse, or balanced flow

The shared model is useful only after each term is tied to measurable proxies and failure conditions. Without those proxies, the mapping remains an analogy.

4 Sequence-Level Lessons

1. Infrastructure and grids show the cost of fixed physical constraints when flow surges faster than capacity can relax.
2. Internet, cloud, and distributed-compute papers show how software systems can trade statefulness for responsiveness.
3. Knowledge graphs and autonomous systems show that informational systems also carry memory, backlog, and routing constraints.
4. Computational architecture shows the same audit pattern at chip and memory-hierarchy scale, where throughput is bounded by locality, power, and translation constraints.

5 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

6 Current Runtime Diagnostic

The release-local discovery script does not simulate every Part B system. It runs a current-runtime domain adapter sweep and a universal network-cascade stress test. The Part B rows in `Part_B_Engineered_Systems/outputs/domain_adapter_sweep.csv` provide a compact audit for energy systems, network routing, cloud capacity, and artificial intelligence. Those rows check finite-value behavior and regime reporting; engineering claims still need engineering measurements.

7 Falsification Targets

The Part B synthesis weakens if an engineered system cannot define measurable throughput and constraint proxies, if those proxies fail to predict overload or recovery behavior better than existing domain models, or if adding CDFD/AFL variables only renames known quantities without improving auditability.

8 Major Universal Upgrade: Mechanism, Scale, and Tests

8.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Engineering Synthesis, the proposed drive is service demand across coupled engineered networks. The active limitation is interdependent capacity, topology, maintenance, and control limits. Responsiveness is carried by redundancy, control, repair, substitution, and coordinated recovery, while retained state is represented by aging, technical debt, design lock-in, and prior failures. The outcome to explain is resilience, cascade containment, service equity, and restoration.

Table 1: Operational CDFD/AFL map for Paper B-12.

Term	Paper-specific observable meaning	Measurement question
Φ	service demand across coupled engineered networks	What rate, load, gradient, or demand enters the focal system?
C	interdependent capacity, topology, maintenance, and control limits	Which measured bottleneck limits transfer, function, or recovery?
S	redundancy, control, repair, substitution, and coordinated recovery	Which process changes effective capacity or reroutes the load?
M_s	aging, technical debt, design lock-in, and prior failures	Which retained state makes equal present loads produce different futures?
Y	resilience, cascade containment, service equity, and restoration	Which outcome is predicted out of sample, with uncertainty?

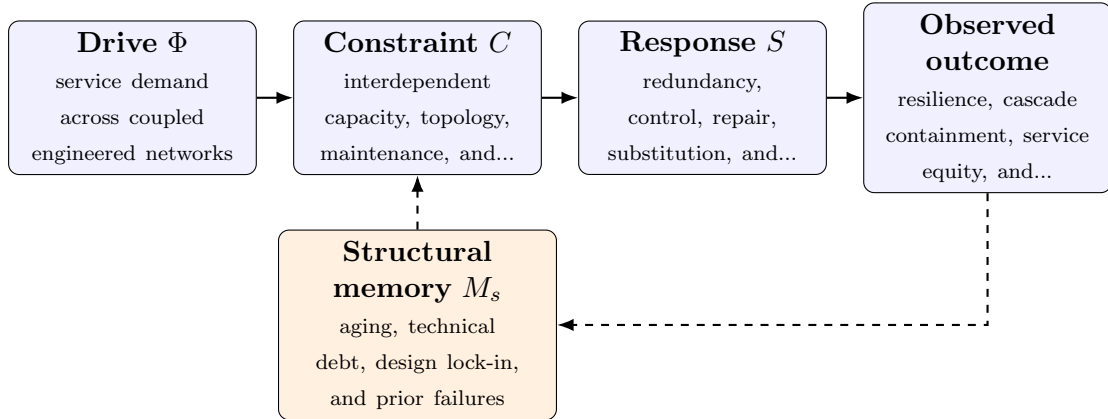


Figure 1: Paper B-12 observable CDFD/AFL architecture for Engineering Synthesis. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

8.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (1)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (2)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (3)$$

$$\Psi_s(t) = \frac{\widehat{\Phi}(t)}{\epsilon + \widehat{C}(t)} \widehat{S}(t) [1 + \omega \widehat{M}_s(t)]. \quad (4)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in service demand across coupled engineered networks arrives faster than redundancy, control, repair, substitution, and coordinated recovery can compensate. This raises or redistributes interdependent capacity, topology, maintenance, and control limits. If the load persists, the retained state encoded by aging, technical debt, design lock-in, and prior failures changes the next response, so a later event of equal magnitude can produce a different trajectory in resilience, cascade containment, service equity, and restoration. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

8.3 Cross-Scale Universality Without Mechanistic Erasure

For the engineered systems family, the universal comparison is between demand, designed capacity, feedback control, dependency topology, and accumulated technical state. The strongest engineering use is prospective: the mapping must identify a bottleneck, a control action, or a recovery signature before the failure occurs. [2, 9, 8, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is microseconds to decades; components to systems of systems. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

Table 2: Evidence ladder for Paper B-12. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure cross-infrastructure telemetry, dependency maps, stress tests, and incident histories and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a common-cause shock across two engineered networks while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: cross-network CDFD terms do not improve cascade or recovery prediction.

8.4 Evidence Ladder and Discriminating Tests

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ , separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

8.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from cross-infrastructure telemetry, dependency maps, stress tests, and incident histories and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of resilience, cascade containment, service equity, and restoration. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

9 Conclusion

The engineered-systems sequence frames infrastructure, computation, networks, and autonomy as systems where flow, constraint, responsiveness, and memory can be audited together. That framing is useful when it produces measurable failure conditions and

reproducible diagnostics. It remains hypothesis-level until it is compared against domain-specific data and engineering baselines.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

10 Reference Layer

References

- [1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: <https://doi.org/10.1103/PhysRevLett.59.381>.
- [2] Albert-Laszlo Barabasi and Reka Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. DOI: <https://doi.org/10.1126/science.286.5439.509>.
- [3] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20250821>.
- [4] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20264779>.
- [5] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20344695>.
- [6] Steve Bico Mujjabi. Cdfd part iv: Universal adaptive flux limitation - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20395901>.
- [7] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfd execution engine, 2026. DOI: <https://doi.org/10.5281/zenodo.20343160>.
- [8] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [9] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of small-world networks. *Nature*, 393:440–442, 1998. DOI: <https://doi.org/10.1038/30918>.